[153] Seat No.______ No. of Printed Pages : 04

SARDAR PATEL UNIVERSITY

B.Sc. (CA & IT) SEMESTER-V EXAMINATION-2019

PS05FIIT01 (OPERATIONS RESEARCH)

November 18, 2019 (Monday)

2.00 p.m. to 5.00 p.m Maximum Marks: 70

Q.1 Choose the correct option in the following multiple choice questions. [10]

1 Operations research is the application of ____________ methods to arrive at the optimal solutions to the problems."

[a] economical [b]scientific

[c]a and b both [d] artistic

2 In graphical representation the bounded region is known as _________ region.

[a] Solution [b]feasible solution [c] basic solution [d] optimal

3 The linear function of variables which is to be maximized or minimized is called _________.

[a]constraints [b]basic requirements

[c]objectivefunction [d]none of them

4 The_________ variable is added to the constraint of less than equal to type.

[a] slack [b] Surplus [c] artificial [d]basic

5 The coefficient of artificial variable in the objective function is ____________.

[a] –M [b] +M [c] 0 [d]none of them

6 In non-degenerate solution number of allocated cell is____________.

[a] Equal to m+n-1 [c] not equal to m+n-1

[c] Equal to m+n-1 [d] not equal to m+n+1

7 North – West corner refers to ____________.

[a] top left corner [b] top right corner [c] both of them [d] none of them

8 The _____ method used to obtain optimum solution of travelling salesman problem.

[a] Simplex [b]Hungarian [c] dominance [d] graphical

9 Activity which does not require any resources or time is called _____.

[a]dummy [b]Predecessor [c] successor [d] none of them

10 Merge event represents ___________ of two or more events.

[a] beginning [b]completion [c] splitting [d] none of them

Q.2 Attempt the short questions: **(any ten)** [12]

1 Define Operation research

2 Define feasible solution

3 Write down any two scopes of operation research.

4 Define surplus variables

5 Define artificial variables

6 Give any four models of operations research

7 Give mathematical form of assignment problem

8 What is transportation problem?

(P.T.O)

9 What is feasible solution and non degenerate solution in transportation problem?
10 What is dummy activity?
11 What is total float?
12 What is free float and independent float?

Q.3

(a) Explain the history of operations research [5]

(b) Give the limitations of operations research [5]

OR

(c) A firm manufactures two types of products A and B and sells them at a profit of Rs. 20 on type A and Rs. 30 on type B. each product is processed on two machines G and H. type A requires 1 minute of processing time on G and 2minutes on H; Type B requires 1 minute on G and 1 minute on H. the machine G is available for not more than 6 hours while H is available for 10 hours during any working day. Formulate this problem as a linear programming problem. [5]

(d) Solve the LPP by graphical method: [5]

Maximize $Z = 3x_1 + 5x_2$

Subject to $x_1 + 2x_2 \le 20,$

$x_1 + x_2 \le 15,$

$x_2 \le 8,$

$x_1, x_2 \ge 0.$

Q.4

(a) Solve the LPP by Simplex method: [5]

Maximize $Z = 3x_1 + 5x_2$

Subject to $x_1 + x_2 \le 4, \quad 3x_1 + 2x_2 \le 18, \quad x_1, x_2 \ge 0$

(b) Solve LPP by Big M Method : [5]

Maximize $Z = 3x_1 - x_2$

Subject to $2x_1 + x_2 \ge 2, \quad x_1 + 3x_2 \le 3, \quad x_1, x_2 \ge 0$

OR

(c) Solve the LPP by Simplex method: [5]

Maximize $Z = 7x_1 + 5x_2$

Subject to $x_1 + 2x_2 \le 6, \quad 4x_1 + 3x_2 \le 12, \quad x_1, x_2 \ge 0$

(d) Solve the LPP : [5]

Maximize $Z = 5x_1 + 7x_2$

Subject to $x_1 + x_2 \le 4, \quad 10x_1 + 7x_2 \le 35, \quad x_1, x_2 \ge 0$

Q.5

(a) Obtain the initial solution to above TP using northwest corner method. [5]

	D_1	D_2	D_3	D_4	Supply
O_1	6	4	1	5	14
O_2	8	9	2	7	16
O_3	4	3	6	2	5
Dem.	6	10	15	4	

(b) Solve the following Assignment Problems. [5]

	I	II	III	IV
1	11	10	18	5
2	14	13	12	19
3	5	3	4	2
4	15	18	17	9

OR

Q. 5 Obtain the Optimal solution using MODI method for the following TP. [10]

	Destination				
Origins	D_1	D_2	D_3	D_4	Supply
O_1	19	30	50	10	7
O_2	70	30	40	60	9
O_3	40	8	70	20	18
Demand	5	8	7	14	

Q.6

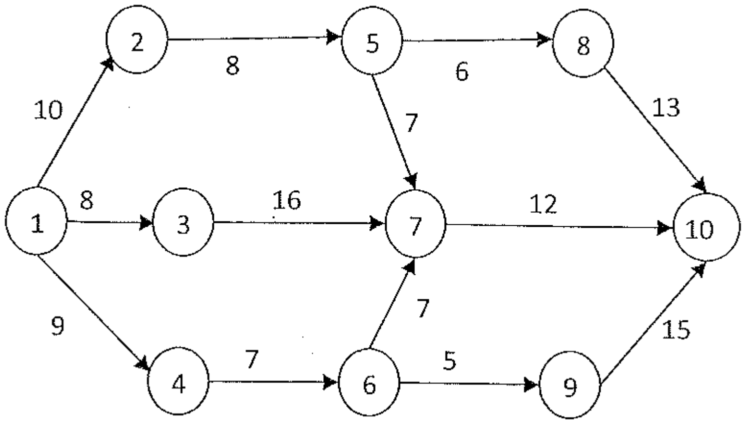
A project has the following time schedule: [10]

Activity	Time In month	Activity	Time In month	Activity	Time In month
1-2	2	3-6	8	6-9	5
1-3	2	3-7	5	7-8	4
1-4	1	4-6	3	7-9	3
2-5	4	5-8	1		

Construct PERT network and compute total float for each activity.
Find Critical path with its duration.

OR

(b) Find the critical path for the following PERT diagram. [4]



(c) In a machine shop 8 different products are being manufactured each requiring time on two different machines A and B are given in the table below: [6]

Product	1	2	3	4	5	6	7	8
Machine-A	30	45	15	20	80	120	65	10
Machine B	20	30	50	35	35	40	50	20

Find an optimal sequence of processing of different product in order to minimize the total manufctured time for all product. Find total ideal time for two machines and elapsed time.

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